Application: SD for Generative Modeling

Learning Generative Models with Sinkhorn Divergences

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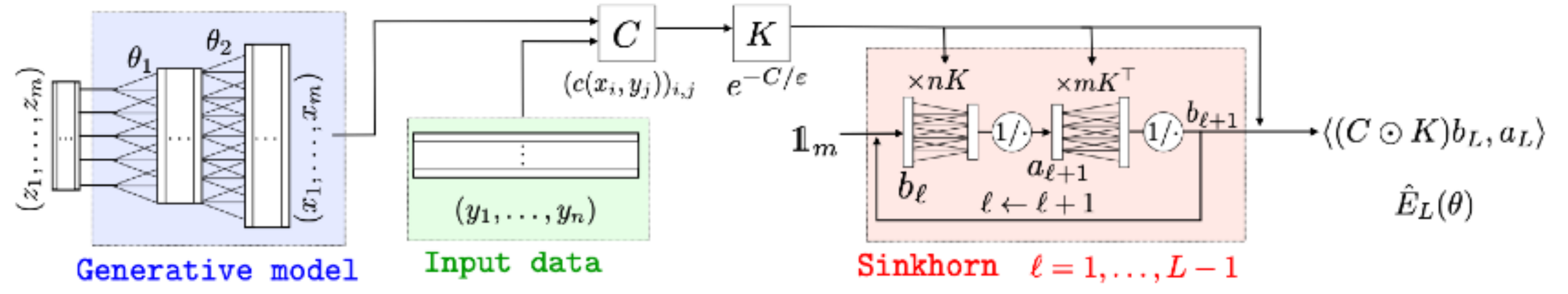
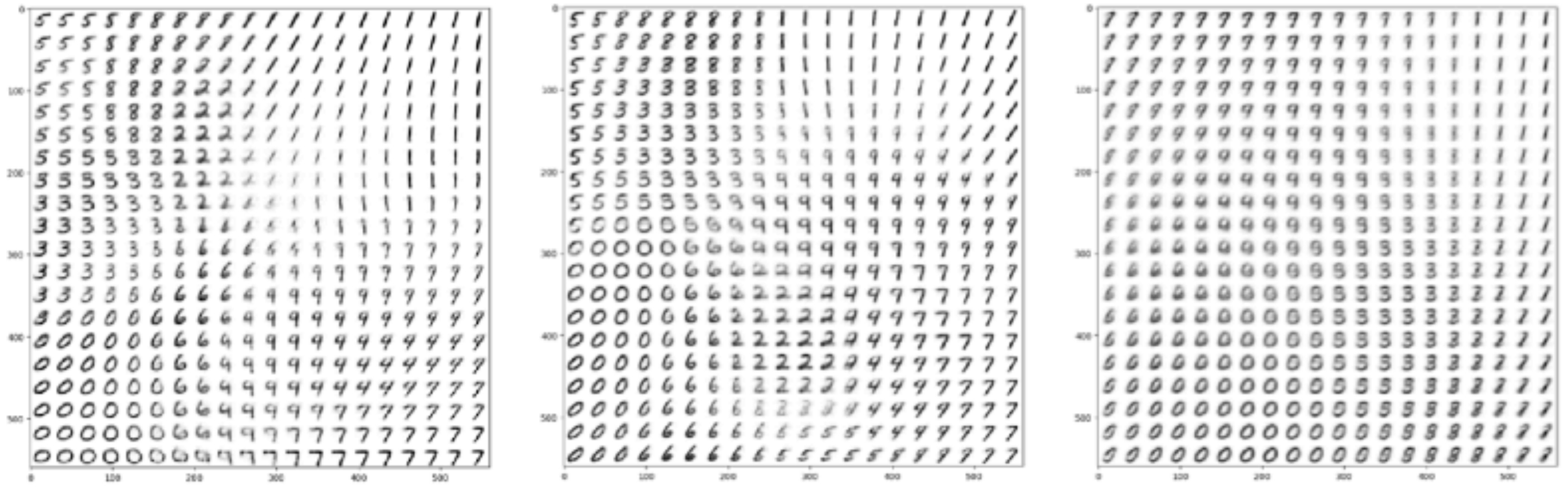
Abstract

The ability to compare two degenerate probability distributions, that is two distributions supported on low-dimensional manifolds in much higher-dimensional spaces, is a crucial

1 Introduction

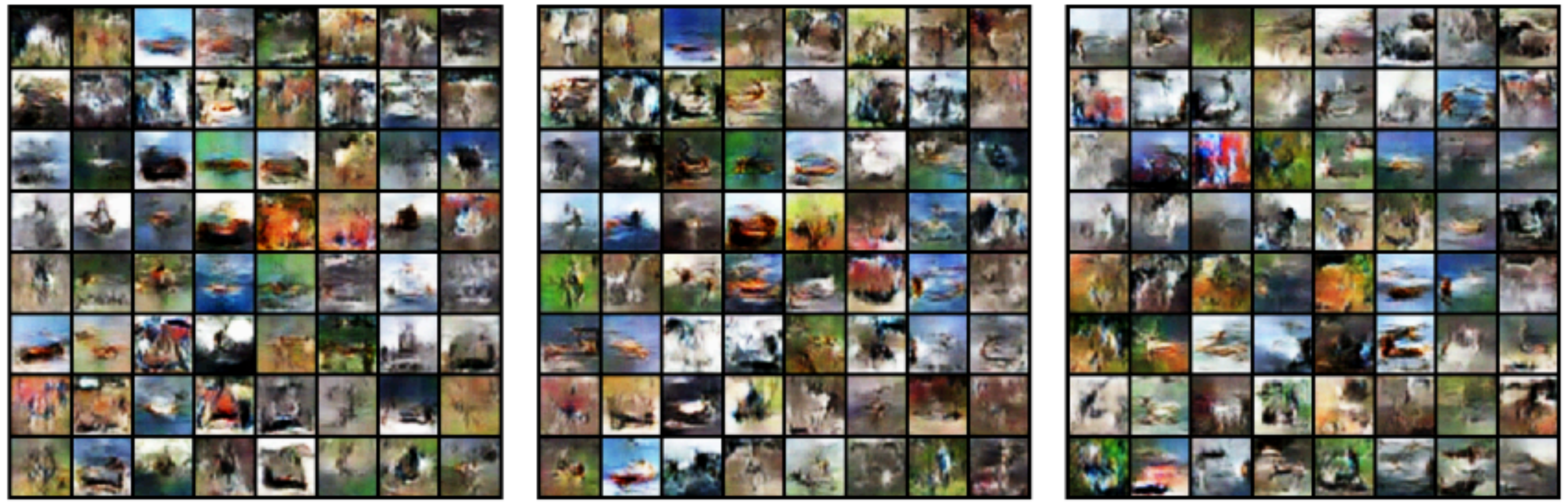
Several important statistical problems boil down to fitting densities, *i.e.* estimating the parameters of a chosen model that *fits* observed data in some meaningful way. While the standard approach is maximum likelihood estimation, this approach is often flawed in





(a) $\varepsilon = 1$, $m = 200, L = 10$ (b) $\varepsilon = 10^{-1}$, $m = 200, L = 100$ (c) $\varepsilon = 10^{-1}$, $m = 10, L = 300$

Figure 3: Influence of the hyperparameters on the manifold of generated digits.



(a) MMD (b) $\varepsilon = 1000$ (c) $\varepsilon = 10$

Part 2: Wasserstein Barycenters